

BIOTECHNOLOGY AND ITS APPLICATIONS

* Harnessing Biology for Human Welfare ~ *

INTRODUCTION

- * Biotechnology is the application of scientific and engineering principles to the processing of materials by biological agents to provide goods and services.
- * It involves techniques that use living organisms or their components to develop useful products and technologies.



DNA
- The genetic material

SCOPE OF BIOTECHNOLOGY

- Improvement in food and agriculture
- Production of medicines and vaccines
- Environmental management
- Industrial products and processes
- Forensic science and diagnostics
- Gene therapy and health care



KEY FEATURES

- Uses living cells or their components
- Interdisciplinary (Biology, Chemistry, Engineering, IT)
- Product oriented
- Ethical and environmental considerations

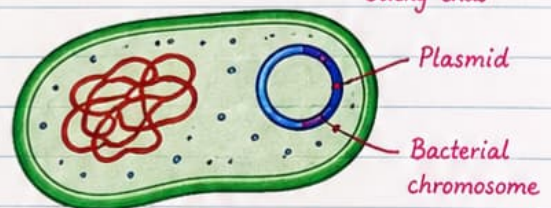


TOOLS OF RECOMBINANT DNA TECHNOLOGY

1. **RESTRICTION ENZYMES** : Act as molecular scissors that cut DNA at specific recognition sequences.
→ Example : EcoRI, HindIII
2. **VECTORS** : DNA molecules used to carry foreign DNA into host cells.
→ Example : Plasmids, Bacteriophages
3. **HOST CELLS** : Cells that accept recombinant DNA and help in its replication.
→ Example : E. coli

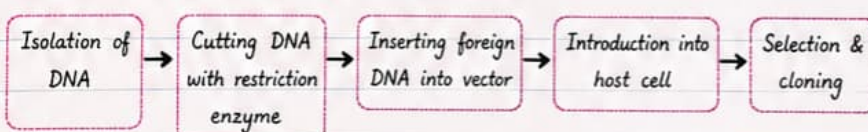


Sticky ends



E. coli

BASIC STEPS IN rDNA TECHNOLOGY



REMEMBER

Biotechnology = Biology
+ Technology
= Better Tomorrow

- * This technology forms the basis of modern biotechnology.

RECOMBINANT DNA (rDNA) TECHNOLOGY (Continued...)

4. CLONING :

Making many identical copies of a recombinant DNA (gene of interest).

→ Example : Multiple copies of human insulin gene.



Bacterial colonies (clones)

Why Cloning?

- To get large amount of DNA
- For further experimentation
- For production of useful proteins

APPLICATIONS OF rDNA TECHNOLOGY

1. Medicine

- * Production of human insulin, growth hormone, interferons, vaccines (e.g., Hepatitis-B vaccine)
- * Gene therapy for genetic disorders.



2. Agriculture

- * Bt crops (e.g., Bt cotton) resistant to insect pests.
- * Improved yield, quality & nutritional value.



3. Environment

- * Bioremediation - using microbes to clean polluted sites (oil spills, toxic wastes).
- * Production of biofuels & bioplastics.



4. Industry

- * Production of enzymes (e.g., lipase, amylase).
- * Fermentation products - vitamins, antibiotics.

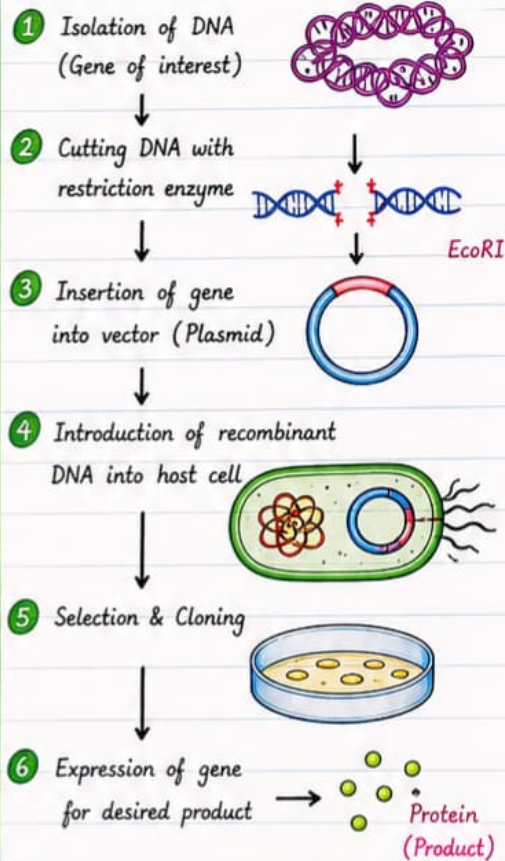


5. Forensic Science

- * DNA fingerprinting for crime detection and paternity testing.



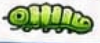
STEPS IN rDNA TECHNOLOGY



IMPORTANT POINT

- * Restriction enzymes act as molecular scissors.
- * Vectors mostly used : Plasmids, Bacteriophages.
- * Bacteria such as *E. coli* are common host cells.

IMPORTANT EXAMPLES OF BIOTECHNOLOGICAL PRODUCTS

PRODUCT	SOURCE (ORGANISM)	TECHNOLOGY USED	USES
Human Insulin	<i>E. coli</i>	rDNA technology	Treatment of Diabetes 
Hepatitis-B Vaccine	Yeast (<i>S. cerevisiae</i>)	rDNA technology	Prevents Hepatitis-B 
Bt Cotton	<i>Bacillus thuringiensis</i> gene	Genetic engineering	Resistant to bollworm 
Interferons	<i>E. coli</i> / Yeast	rDNA technology	Treatment of viral infections 
Streptokinase	<i>Streptococcus</i>	Fermentation	Clots dissolution in heart attack 
Ethanol (Biofuel)	Yeast	Fermentation	Fuel, Solvent 

REMEMBER ★

Biotechnology improves quality of life by combining Biology + Technology for the welfare of humanity. ♥



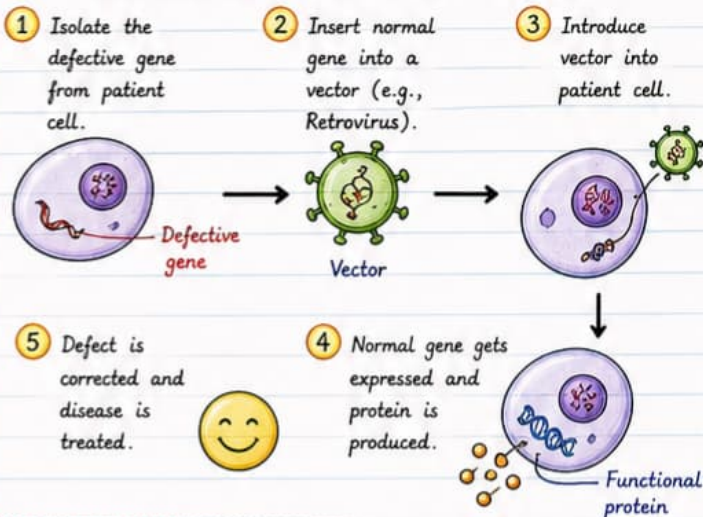
GENE THERAPY

* Gene therapy is a technique to treat genetic disorders by inserting a **normal and functional gene** into the patient's cells.

* It can be of two types :

1. **Somatic gene therapy** - Gene is transferred to somatic (body) cells; the changes are not inherited.
2. **Germ line gene therapy** - Gene is transferred to gametes or zygote; the changes are inherited by next generation.

STEPS OF GENE THERAPY (SOMATIC)



TRANSGENIC ANIMALS

- * Animals that carry a foreign gene are called **transgenic animals**.
- * The foreign gene may be from the same species or from a different species.
- * These animals are useful for studying gene function and for producing valuable products.

EXAMPLES

- Transgenic mice with cancer genes - to study cancer.
- Transgenic cows with human α -lactalbumin gene - produces protein in milk (used in infant food).
- Transgenic sheep - "Pharming" animals that produce human proteins in their milk.



BIOTECHNOLOGY AND AGRICULTURE

1. Bt CROPS (GENETICALLY MODIFIED CROPS)

- * Bt gene is taken from *Bacillus thuringiensis* which produces a protein (Cry protein) toxic to certain insects.
- * Bt gene is inserted into crop plants to make them insect resistant.

Example : Bt Cotton, Bt Brinjal, Bt Maize



Bt Cotton

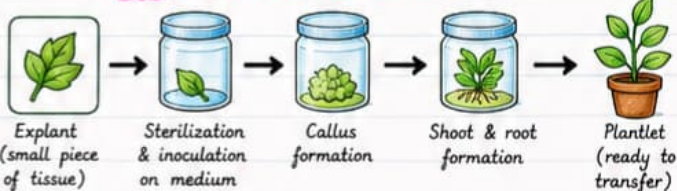
BENEFITS

- ✓ Reduces crop loss due to insects.
- ✓ Less use of chemical pesticides.
- ✓ Increases yield and quality.

2. TISSUE CULTURE (MICROPROPAGATION)

- * Technique to grow a large number of identical plants from a small piece of plant tissue.
- * Used for plants that are difficult to propagate by seeds.

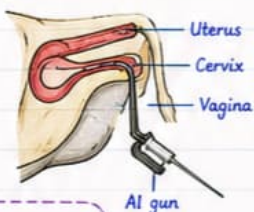
STEPS IN TISSUE CULTURE



BIOTECHNOLOGY IN ANIMAL HUSBANDRY

1. ARTIFICIAL INSEMINATION (AI)

- * Semen is collected from superior male and artificially introduced into the female reproductive tract.
- * Improves breed, high milk yield, disease control.



2. ANIMAL BIOTECHNOLOGY

- * Production of vaccines, hormones and enzymes using animal cell culture.
- Example : Production of interferons, clotting factors etc.



QUICK REVISION

- ✓ Biotechnology = Biology + Technology
- ✓ It uses living systems for useful products.
- ✓ rDNA technology is the backbone of modern biotechnology.
- ✓ Applications are in medicine, agriculture, environment, industry and animal husbandry.

KEY TAKEAWAY

Biotechnology helps in improving the quality of life, conserving environment and solving problems of health, food and industry. ♥

From tiny genes to big solutions - Biotechnology builds a better tomorrow.

BIOTECHNOLOGY AND MEDICINE

- * Biotechnology has revolutionized the medical field by providing new diagnostics, drugs and vaccines.
- * Recombinant DNA technology is used to produce therapeutic proteins.
- * Monoclonal antibodies are used for detection and treatment of diseases like cancer.
- * Gene therapy shows promise for treating genetic disorders.

EXAMPLES OF BIOTECH PRODUCTS IN MEDICINE

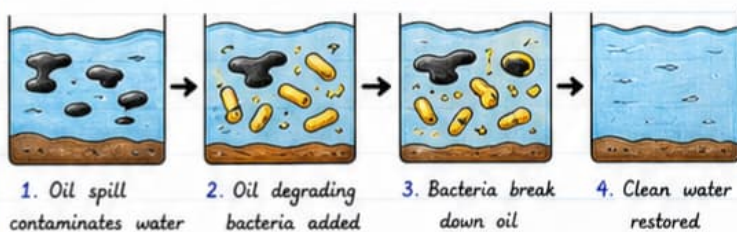
PRODUCT	SOURCE/TECHNOLOGY	USES
Human Insulin	rDNA (E.coli / Yeast)	Treatment of Diabetes 
Hepatitis-B Vaccine	Recombinant yeast	Prevents Hepatitis-B 
Interferons	rDNA technology	Treatment of viral infections 
Monoclonal Antibodies	Hybridoma technology	Cancer therapy, diagnostics 
Streptokinase	Streptococcus (fermentation)	Clot buster in heart attack 

BIOTECHNOLOGY AND ENVIRONMENT

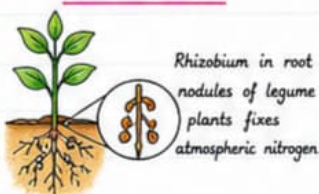
APPLICATIONS

- **Bioremediation** - Use of microbes to degrade pollutants (oil spills, pesticides, heavy metals).
- **Sewage treatment** - Microbes break down organic waste in sewage.
- **Biofertilizers** - Microbes like Rhizobium, Azotobacter fix nitrogen and improve soil fertility.
- **Biopesticides** - Natural agents to control pests (e.g., Bt toxin from Bacillus thuringiensis).

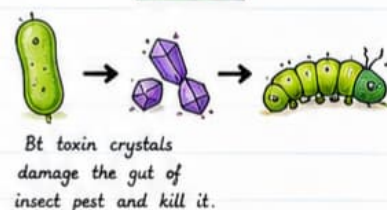
BIOREMEDIATION OF OIL SPILL



BIOFERTILIZERS



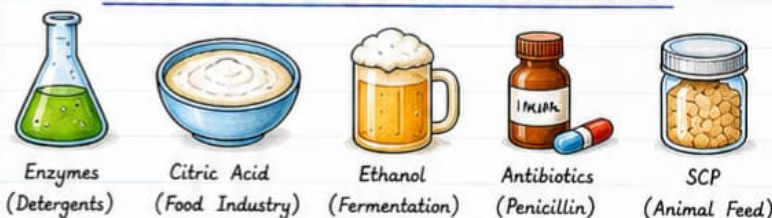
BT TOXIN



BIOTECHNOLOGY AND INDUSTRY

- * Used in production of enzymes (amylase, lipase, protease) for detergents, food, leather and textile industries.
- * Fermentation is used to produce antibiotics, organic acids, vitamins, alcohol and other chemicals.
- * Single cell protein (SCP) is produced from microbes and used as animal feed.

INDUSTRIAL PRODUCTS THROUGH BIOTECHNOLOGY



ETHICAL AND SAFETY CONCERNS



- Ethical issues in gene cloning and genetic modification.
- Possible harm to non-target organisms and biodiversity.
- Safety of GM foods and long term health effects.
- Need for strict regulation and public awareness.



SUMMARY

- ✓ Biotechnology uses living systems for human welfare.
- ✓ It has wide applications in medicine, agriculture, environment and industry.
- ✓ rDNA technology is the key tool behind many biotechnological advances.
- ✓ Biotechnology offers solutions to global challenges and improves quality of life.

